ROAD READER · PROCESSING ALGORITHM

# How the box reads the road

From handlebar vibration to a map of road condition and potholes

THE CORE IDEA

# The box feels shaking, not the road

**Road**

Bumps, cracks and potholes

**Bike reacts**

Tyres squash, frame and rider bounce

**Box measures**

Push and rotation, 200 times a second

**The software runs this backwards.** It knows how a bike reacts to a bump, so it works out which road would have caused the shaking it measured.

THE PIPELINE

# Five steps from recording to map

1

### Tidy up

Line readings up on a steady beat and mark lost Bluetooth packets

2

### Find the mount

Standing still shows which way is down; speeding up shows forward

3

### Remove the riding

Track tilt and speed, then subtract gravity, braking and turning

4

### Rebuild the road

Run the bike’s bounce in reverse to get the road’s shape (the core)

5

### Grade and pin

Score every 20 m and mark potholes and bumps on the map

**IN**

Accelerometer, gyroscope and the phone’s GPS

**OUT**

Road shape every 5 cm, condition every 20 m, pothole pins

STEPS 1–3

# Every push the box feels has a cause

| The box feels… | Caused by | What the software does |
|---|---|---|
| A steady 1 g pull | Gravity | Standing still for 2 s, learns which way is down |
| A push as GPS speeds up | Pedalling | Learns which way is forward |
| A forward push | Braking, not a downhill | Subtracts it, using GPS speed changes |
| A sideways push | Leaning into a turn | Subtracts it, using GPS speed × turn rate |
| A slow nose-up tilt | A hill | Adds it up over distance for the hill profile |
| Fast up-and-down shaking | The road | Keeps it and sends it to step 4 |

The box can be mounted at any angle: it works out “down” and “forward” for itself.

STEP 4 · THE CORE

# Running the bike in reverse

1 Front wheel hits the bump → 2 Frame and rider bounce → 3 Rear wheel hits it a moment later

### Two clues, not one

Up-and-down shaking plus forward-and-back rocking. Front and rear hits rock the bike in opposite directions.

### Don’t trust every wiggle

If a wiggle is more likely rider movement or sensor noise than road, it’s left out instead of becoming a bump.

### Four-second chunks

Speed barely changes in 4 s, and speed decides how quickly the bumps arrive.

### Gaps stay blank

Lost Bluetooth readings count as “nothing happened”. Guessing across a gap would invent potholes.

STEP 5

# Grading the road, pinning potholes

## A condition score every 20 m

The International Roughness Index city road departments use: how much a standard car’s suspension would move at 80 km/h.

| IRI (m/km) | Label |
|---|---|
| Under 2 | Good |
| 2 to 4 | Fair |
| 4 to 6 | Poor |
| 6 and up | Very poor |

## Potholes and bumps

- Dips and rises that stand out from the road around them
- Two sizes: 10 cm–2 m (potholes, plates) and 1–8 m (sunken patches, speed humps)
- At least 20 mm, and a higher bar on roads that are rough everywhere
- Duplicates and echoes beside sharp potholes are removed
- Each pin gets a GPS location, a time and a severity from 0 to 1

HOW WELL IT WORKS · SIMULATED RIDES

# Strong on roughness, rough on hills

**0.85–0.99**
match to the true road shape (1 = perfect)

**1–1.5 m/km**
typical error in the condition score

**1 per 350 m**
false alarms on a heavily potholed street

**60–80%**
of true depth: use it to rank potholes

Hills on a bike are approximate: 0.5–1 m off per km, up to 3 m with lots of turns.